

GEOGRAPHY (H.C.G. PAPER – 2)

PART I (30 Marks)

(Attempt *all* questions from this *Part*.)

Question 1

Study the extract of the **Survey of India** Map Sheet No. **G43S7** and answer the following questions:

- (i) (a) Give the **four-figure** grid reference for $\Delta 224$. [2]
(b) What is the **contour interval** used in the given map extract?
- (ii) (a) Mention two man made features seen in **grid square 2706**. [2]
(b) Mention two natural features seen in **grid square 2905**.
- (iii) 1) What do you understand by **1:50,000** printed below the map extract? [2]
2) What is the *pattern of drainage* seen in the **grid square 2505**?
- (iv) (a) What do you understand by **.10r in 2709**? [2]
(b) What is the *compass direction* of **Mahudi Nani (2604)** from **Panswala (2909)**?
- (v) Calculate the **area** in *kilometre square* of the region enclosed between **Easting 26 to 29** and **Northing 05 to 11**. [2]

Comments of Examiners

- (i) (a) Majority of the candidates answered it correctly. In some of the cases, candidates mentioned 'northing' prior to 'easting' number.
- (b) Some of the candidates defined the term instead of mentioning the 'actual contour interval'. 'Unit' was not mentioned by some of the candidates.
- (ii) (a) Most of the candidates answered it correctly. In some cases, candidates mentioned 'overhead tank', but no symbol of 'overhead tank' was found in this 'grid square'.
- (b) This sub-part was answered correctly by majority of the candidates. Some candidates mentioned 'trees' as natural feature which is incorrect and 'river' instead of 'stream'.
- (iii) 1) Some candidates explained it in terms of 'units' which was incorrect as 'Representative fraction' does not have unit.
- 2) It was answered correctly by most of the candidates. Some candidates mentioned 'trellised' instead of 'dendritic'.
- (iv) (a) Some candidates did not write the correct term – 'Relative height'. Moreover, some candidates incorrectly mentioned '10 meters above mean sea level'.
- (b) It was answered correctly by several candidates.
- (v) Some candidates did not calculate the answer correctly and other candidates failed to mention the 'unit'.

Suggestions for teachers

- Provide regular practice regarding four-figure reference where easting number should be written before northing number. It should be left side easting and lower side northing.
- Guide students to mention 'unit'.
- Familiarise students with the 'index' given below the map and all symbols in the index.
- Explain difference between 'River' and 'Stream' to differentiate them on the map.
- Encourage students to look at the 'pattern seen in the square' that is referred to rather than looking in 'wider adjacent area'.
- Diagrams may be used to explain 'different types of drainage patterns'.
- Enough identification practice should be done using the 'toposheet'.
- Guide students on the following:
 - any number with small 'r' depicts 'relative height' which is in relation to the 'surrounding area' and not 'sea level'.
 - 'Height above sea level' is shown by 'contours', 'spot height', and 'benchmark' and 'triangulated station'.
 - familiarize students with the terms – 'Height'; 'Relative Height' and 'Altitude'.
- Discuss '8 cardinal points of the compass'.
- Enable students to do 'simple calculation of area' by using 'scale of the map'.
- Encourage students to mention the 'unit' after the answer (as asked in the question paper).

MARKING SCHEME		
Question 1		
(i)	(a)	2706
	(b)	20 metres
(ii)	(a)	Man made (2706) – lime kiln, covered tank, lined well, permanent hut/ hut/ settlement/ powerline/ powerline with poles/ <i>chattri</i> / metalled road/ well. <i>(Any two points)</i>
	(b)	Natural (2905) – dry stream or stream/ seasonal stream/ sand dune/ sandy feature/ sand hill/ broken ground/ barren land/ plain. <i>(Any two points)</i>
(iii)	(a)	It is Representative Fraction/RW/ type of scale/ scale in the form of ratio/ definition of RF/ it is universal scale.
	(b)	Dendritic
(iv)	(a)	It is the relative height of sand dune which is 10 metres.
	(b)	Southwest/ SW
(v)	Area = Length × Breadth = Scale of the given map is 2 cm to 1 km. = 3 × 6 = 18 grid square Area of 1 grid square - 1 km ² ∴ 18 km ²	

Question 2

On the outline map of India provided:

- | | | |
|-------|--|-----|
| (i) | Mark and label the <i>Satpura</i> . | [1] |
| (ii) | Shade and label the <i>Gulf of Kutch</i> . | [1] |
| (iii) | Mark and label the <i>River Chambal</i> . | [1] |
| (iv) | Mark and label <i>Delhi</i> . | [1] |
| (v) | Mark and label <i>80°30' E longitude</i> . | [1] |
| (vi) | Mark and label <i>Jharia coalfield</i> . | [1] |

- (vii) Mark with arrows and name the *Southwest Monsoon winds*. [1]
- (viii) Mark and label the *Karakoram Pass*. [1]
- (ix) Shade and label a *sparsely populated region in Western India*. [1]
- (x) Shade an area having *Alluvial soil*. [1]

Comments of Examiners

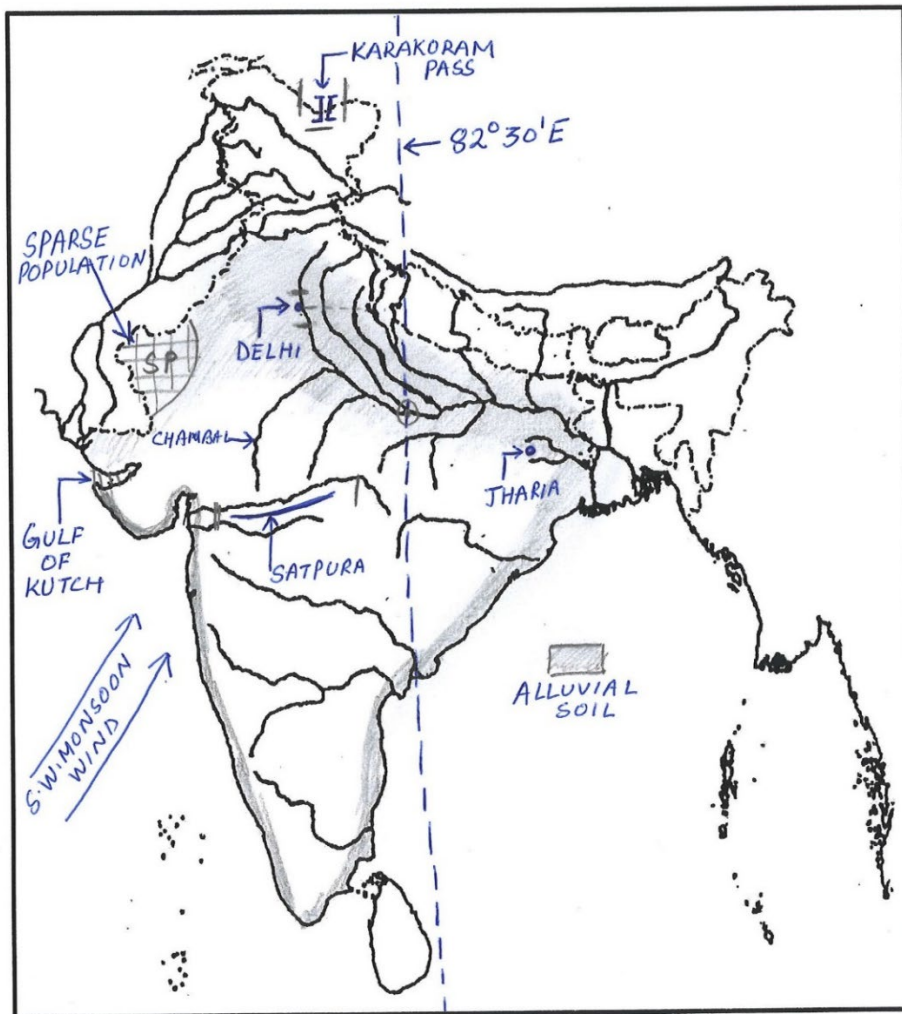
- (i) Some candidates made a line too broad, and it touched the rivers on both sides or one side.
- (ii) Several candidates marked in area of 'Gulf of Khambat' instead of 'Gulf of Kutch'.
- (iii) 'River Chambal' was marked and labelled correctly by majority of the candidates. Some candidates mentioned, 'Chambal' in between the 'rivers' without the use of arrow pointing towards the right direction.
- (iv) Some candidates made very large dot, and marked on the left bank of Yamuna. In some cases, candidates made dot away from Yamuna.
- (v) Most of the candidates marked it correctly while few of them marked 'Tropic of Cancer' instead.
- (vi) A very large area was marked by some of the candidates.
- (vii) It was marked correctly by several candidates. In some cases, the candidates missed making arrowhead, however, few of them made the arrows in incorrect direction.
- (viii) Some candidates incorrectly marked 'Nathu La' or made the pass towards 'west ward'.
- (ix) It was marked correctly, however, some candidates did not label or marked it in the 'eastern side' rather than 'western side'.
- (x) It was marked correctly. Some candidates either missed 'labelling' or 'shaded the areas of Nepal' which was incorrect.

Suggestions for teachers

- Mark Satpura between Narmada and Tapi (Tapti) and not touch the river (should be made parallel).
- Provide regular practice of map work and emphasize the following points:-
 - use of arrows and shading for marking the Gulfs and Bays.
 - show arrows to point towards the correct answer. Guide students to shade/highlight the river using blue colour.
 - 'Delhi is on the right bank of Yamuna' and so the dot should be made touching the river.
 - clarify the difference between 'latitude and longitude'.
 - 'Jharia' should be marked 'between the fork' of river 'Damodar'.
 - arrows should be 'made in water body' pointing 'towards the land' and label it.
 - advise students to read the question properly.
 - guide students to do the labelling after marking/shading the required area.
- Provide clarity in directions like East, West, North, or South to the student.
- Encourage students to do shading only in 'Indian' area and not in the 'neighbouring countries'.

MARKING SCHEME

Question 2



Question 3

[10]

Choose the correct answers to the questions from the given options.

(Do not copy the questions, write the correct answers only.)

(i) Which of the following area *receives rain* from the *North East Monsoon*?

- (a) Konkan coast
- (b) Ganga basin
- (c) Coromandel coast
- (d) Malabar coast

(ii) Which is the **most widespread soil** of India?

- (a) Red soil
- (b) Alluvial soil
- (c) Laterite soil
- (d) Black soil

(iii) Which type of forest is found **mostly** in the *coastal areas of India*?

- (a) Tropical Evergreen Forest
- (b) Tropical Desert Forest
- (c) Littoral Forest
- (d) Tropical Deciduous Forest

(iv) Which of the following is a **ground water source**?

- (a) Pond
- (b) Lake
- (c) Spring
- (d) River

(v) **Copper** is alloyed with **Zinc** to form:

- (a) Stainless steel
- (b) Brass
- (c) Bronze
- (d) Aluminium

(vi) **Hirakud dam** is based on which of the following *rivers*?

- (a) River Godavari
- (b) River Mahanadi

- (c) River Krishna
- (d) River Narmada
- (vii) Which of the following is **NOT** a problem of Indian agriculture?
 - (a) Dependence on monsoon
 - (b) Small land holdings
 - (c) Two main cropping seasons
 - (d) Use of traditional methods of farming
- (viii) Which of the following is a basic / key industry?
 - (a) Iron and Steel Industry
 - (b) Silk Industry
 - (c) Electronic Industry
 - (d) Cotton textile Industry
- (ix) Which of the following is a staple food grain of Indians?
 - (a) Wheat
 - (b) Rice
 - (c) Gram
 - (d) Arhar
- (x) Which of the following method of waste disposal is *harmful*?
 - (a) Composting
 - (b) Segregation
 - (c) Dumping
 - (d) Vermicomposting

Comments of Examiners

- (i) Some candidates mentioned 'Malabar coast' rather than 'Coromandel coast'.
- (ii) Majority of the candidates answered it correctly.
- (iii) Some candidates missed reading 'mostly', thus, mentioned 'Tropical Evergreen' instead of 'Littoral'.
- (iv) It was answered correctly by majority of the candidates, however, some mentioned 'Pond' as ground water source.
- (v) Candidates were confused between 'brass and bronze' as some candidates did not understand the term 'alloy'; and mentioned 'aluminium'.
- (vi) It was answered correctly by majority of candidates, however, some mentioned 'Narmada' which was incorrect.
- (vii) Some candidates mentioned the 'unique feature of Indian agriculture' as response instead of the 'problem' which the question demanded.
- (viii) Most of the candidates answered it correctly. Some candidates mentioned 'cotton textile' indicated the lack of clarity on the term 'key industry'.
- (ix) Majority of the candidates responded it correctly as they mentioned 'Wheat and Rice'.
- (x) It was answered correctly by many candidates, however, in certain cases, candidates mentioned 'Vermicomposting' reflecting lack of clarity on the concept.

Suggestions for teachers

- Work on the conceptual clarity of the topic 'climate' and ensure students know the difference between S. W. Monsoon and N. E. Monsoon.
- Guide students to read the question and options given therein properly to answer the question correctly.
- Familiarize students with the terms 'ground water' and 'surface water' with examples.
- Clarify uses of the minerals.
- Familiarize with the 'rivers on which the dams' are made.
- Students must know the meaning of a 'basic' or 'key industry'.
- Focus on the concept 'Staple grain of Indians'.
- Build clarity of concept related to method of waste disposal and vermicomposting etc.

MARKING SCHEME

Question 3

(i)	(c) Coromandel coast
(ii)	(b) Alluvial soil
(iii)	(c) Littoral forest
(iv)	(c) Spring
(v)	(b) Brass
(vi)	(b) Mahanadi
(vii)	(b) Two main cropping seasons

(viii)	(a) Iron and steel
(ix)	(b) Rice/ wheat
(x)	(c) Dumping

PART II (50 Marks)

(Attempt *any five* questions from this *Part.*)

Question 4

(i) Name the *seasons* of India and mention the *duration* of each season. [2]

(ii) Why maximum rainfall in India occurs after the summer season? [2]

(iii) Give a geographical reason for each of the following: [3]

(a) The *Malabar coast* receives heavy rainfall in the month of *July* but the *Coromandel coast* during this time remains comparatively dry.

(b) *Punjab* receives rainfall in winter season.

(c) *Mumbai* is warmer than *Delhi* in the month of December.

(iv) Study the climate data of Station **X** and answer the questions that follow: [3]

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Temp in °C	21.0	21.9	24.3	27.2	28.0	26.4	26.1	25.4	26.0	26.0	23.8	21.2
Rainfall in cm	5.1	2.8	1.2	1.7	3.9	4.6	8.4	11.4	11.9	31.6	34.5	14.8

(a) Calculate the annual range of temperature of Station **X**.

(b) Name the wind that brings most of the rainfall to this station.

(c) On which coast of India is Station **X** located?

Comments of Examiners

- (i) The question was answered correctly; however, many candidates' responses lacked the technical terms that were expected in the answer.
- (ii) Candidates were not able to explain 'the relation of high heat during summer' leading to 'low pressure conditions on land' which in turn attracts 'the rain bearing winds'.
- (iii) (a) It was answered correctly by majority of the candidates.
 (b) Most of the candidates answered the question correctly. In some cases, candidates mentioned 'source as N E Monsoon' which was an incorrect response.
 (c) Majority of the candidates answered this part correctly. In some cases, candidates mentioned 'nearness to Equator' rather than 'distance from sea'.
- (iv) (a) In some cases, the candidates failed to mention the 'units', however in certain cases, 'calculation errors' were found.
 (b) Most of the candidates answered this question correctly.
 (c) It was answered correctly in most of the cases. Some candidates mentioned 'city instead of coast'. There were few candidates who were not able to identify the coastal area.

Suggestions for teachers

- Guide students to use technical terms relating to the 'seasons such as – hot and dry season, hot and wet season, retreating monsoon season, cold and dry season'.
- Explain and discuss with students the basic concepts related to climate such as concept of 'low pressure' and 'high pressure'.
- 'Relief rainfall' should be explained using diagrams and examples from India.
- Encourage students to provide detailed explanations for both cities when the question mentions two cities.
- Lay emphasis on writing of 'unit' in the calculation related questions.
- Guide students to read question carefully and answer accordingly.
- Discuss the difference between - region/coast/place.
- Regularly provide practice of application-based questions on climate data.

MARKING SCHEME

Question 4

(i)	Summer – hot & dry season/ hot season/ hot weather season Rainy – hot and wet season/ SW monsoon season/ advancing monsoon season/ onset of monsoon/ monsoon. Retreating monsoon Winter – cold & dry season/ NE monsoon season/ cold season/ cold weather season
(ii)	During summer low pressure is created over land. This attracts the wind from sea leading to rain.

(iii)	(a) Coromandel coast is in Tamil Nadu which lies in the rain shadow region of Arabian sea branch and the Bay of Bengal branch blows parallel to the coast. (b) Punjab receives rainfall in winter due to the Western disturbance/ temperate cyclone/ Westerly depression. (c) Mumbai lies near the coast so gets influence by the sea and experiences a moderate climate whereas Delhi has continental location and has extreme temperature conditions resulting in high annual range of temperature.
(iv)	(a) $28^{\circ} - 21^{\circ} = 7^{\circ}\text{C} / 7^{\circ}\text{C}$ (b) Retreating monsoon winds/ NE monsoon (c) Eastern coast/ SE coast/ coromandel coast

Question 5

- (i) (a) Mention *any two* causes of soil erosion in India. [2]
- (b) Name *two* soil conservation methods that may be practised in arid (dry) regions.
- (ii) Mention two characteristics of *Black soil*. [2]
- (iii) Name the soil: [3]
- (a) That is found on the summits of Eastern and Western Ghats.
- (b) That is suitable for dry farming.
- (c) That varies in texture.
- (iv) Give a geographical reason for each of the following: [3]
- (a) *Terrace farming* is an ideal soil conservation method for hilly areas.
- (b) *Khadar* soils are preferred over *Bhangar* soils.
- (c) Foothills of Himalayas have been affected by soil erosion.

Comments of Examiners

- (i) (a) Some candidates mentioned ‘agents of soil erosion’ in place of ‘causes’.
- (b) In some cases, candidates mentioned the ‘conservation methods’ in general instead of writing specific words for ‘arid areas’, however, some candidates did not understand the meaning of ‘arid’.
- (ii) It was answered correctly by several candidates.
- (iii) (a) Candidates were found to be confused related to Red soil/ Black soil.
- (b) Some confusion with ‘Laterite soil’ was found.
- (c) Candidates were not clear on the meaning of ‘texture’.
- (iv) (a) Candidates were unable to explain the ‘steep slope with speed of running water’ and ‘soil erosion’.
- (b) Most of the candidates were able to answer it correctly, however, some candidates were unable to explain ‘Khadar and Bhangar difference’.
- (c) Candidates were not able to describe ‘deforestation’ / ‘heavy rainfall’ to ‘soil erosion’.

Suggestions for teachers

- Share with students the difference between the causes and the agents of soil erosion.
- Make clear to students the conservation methods specific for humid and dry areas and right words to describe about it.
- Discuss composition/characteristics of each type of soil as per the scope of syllabus.
- Explanation regarding formation of each type of soil must be done using a map to show the area in which the soil has developed.
- Focus on characteristics of soil with reference to other types of soil to enable the students to bring out the difference between each type.
- Certain terms such as texture, soil profile, soil conservation etc. must be discussed in class.
- More of concept building is required.
- Explain ‘Khadar and Bhangar difference using map of India’ for better clarity of the concept.
- Share reasons for ‘soil erosion in different areas of India’ and focus on concept building.

MARKING SCHEME

Question 5

(i)	(a) Deforestation, Overgrazing, Faulty farming techniques, urbanisation, industrialisation, lack of vegetation cover, floods, heavy rain, mining, construction of dam, excessive use of chemicals. (b) Planting of shelter belts, strip cropping, afforestation or wind breakers.
(ii)	Sticky when wet, cracks when dry, moisture retentive, black in colour. Colour varies from black to chestnut brown, self-ploughing capacity, fine grained, clayey in texture, rich in LIMCAP, poor in nitrogen, organic matter & phosphorus.
(iii)	(a) Laterite soil (b) Red soil/ Black soil (c) Alluvial soil
(iv)	(a) Terrace farming prevents soil erosion by slowing the speed of running water and thus contributes to soil conservation. (b) Khadar soils is more fertile than Bhangar soils. Khadar is new alluvium, fine grained. Does not have calcite modules. Replenished. (c) Due to large scale deforestation in past/ heavy rain/ torrential downpour

Question 6

- (i) State *two* ways in which forests help in protecting the environment. [2]
- (ii) Mention *two* ways in which Tropical Desert vegetation has adapted to its environment. [2]
- (iii) (a) In which natural vegetation belt is *Sundari* tree found? [3]
- (b) Why is the *Tropical Deciduous Forest* commercially the most exploited forest belt of India? (*Two points*)
- (iv) (a) Name one region in India where Tropical Evergreen Forest is found. [3]
- (b) Mention the *temperature* and *rainfall* conditions found suitable for its growth.

Comments of Examiners

- (i) This question was well answered by most candidates. Some candidates mentioned the importance of forest not in context of /related to the environment.
- (ii) Majority of candidates answered this question correctly.
- (iii) (a) Most of the candidates answered this question in a correct manner. Some candidates mentioned Sundarbans instead of the name of natural vegetation belt (Littoral).
(b) It was answered correctly by the candidates.
- (iv) (a) Candidates were able to answer this question correctly, however, some candidates mentioned the 'name of the states' instead of 'region'.
(b) Some candidates were found confused with the 'temperature and rainfall' and provided incorrect answers. A number of candidates mentioned 'characteristics' rather than 'climatic conditions'.

Suggestions for teachers

- Share importance/benefits of forest for both - economic and environmental aspect.
- Explain, using map of annual rainfall, soil distribution and temperature condition, the growth, and the characteristics of vegetation belts of India.
- Encourage students to learn the names of important trees that grow in the 'natural vegetation belts of India'.
- Revise the characteristics of each natural vegetation belt.
- Explain to students that when the question demands 'region' then merely 'names of the states' is not sufficient.
- Provide regular practice to ensure learning the temperature and rainfall conditions.
- Provide clarity to students regarding appropriate places to mention climatic conditions and its characteristics.

MARKING SCHEME

Question 6

(i)	Moderates the climate, controls humidity, temperature, and precipitation, plays a major role in carbon cycle, absorbs carbon di-oxide, maintain purity of air, controls atmospheric pollution, controls soil erosion, Controls flood, Helps to increase underground water table, helps in percolation, provides humus to soil and increases the fertility, regulates water cycle, reduces the impact of global warming. <i>(Any two points)</i>
(ii)	Xerophytic, leaves modified to thorns, long roots, thick stem, symbiotic relation with stinging ants/ very small leaves/ waxy leaves/ thick leaves/ rolled up leaves/ sunken stomata/ succulent plant/ drought tolerant/ thick stem/ woody stem. <i>(Any two points)</i>

(iii)	(a) Littoral/ Tidal forests/ mangrove (b) sparsely located/scattered-easy to cut/exploit, hard wood-durable, high value of timber, provides fruits and other forest products, easily accessible, easy to transport. (Any two points)
(iv)	(a) Tropical Evergreen Forest is found on the western side of Western Ghats/ western slopes of Western ghats/ windward side of Western ghats/ NE hills/ submountain of West Bengal/ Andaman & Nicobar/ Lakshadweep. Western Maharashtra, Western Karnataka, Kerala, Western Ghats. (b) Temperature – 25° C- 27° C (15 – 30 °C), Rainfall – more than 200 cms.

Question 7

- (i) Give a reason for each of the following: [2]
- (a) Agriculture sector in India needs the support of irrigation.
- (b) *Well irrigation* is a popular means of irrigation in *Northern India*.
- (ii) Distinguish between *Inundation canal* and *Perennial canal*. [2]
- (iii) (a) Where in India is *Tank irrigation* most widely used? [3]
- (b) Give *one* factor that has led to its popularity in the region mentioned by you.
- (c) Mention *one* advantage of using tank as a means of irrigation.
- (iv) (a) What is *Rainwater Harvesting*? [3]
- (b) Give *two* ways in which *Rainwater Harvesting* is important for India.

Comments of Examiners

- (i) (a) Most of the candidates were able to answer it correctly.
- (b) It was well answered by several candidates.
- (ii) Majority of candidates were able to answer it correctly.
- (iii) (a) 'Peninsular region of India/ southern region' was the expected answer, but some candidates mentioned the 'names of the states'.
- (b) It was a linked question, so if the first answer was wrong, then the second one was automatically considered incorrect.
- (c) Most of the candidates responded this question correctly.
- (iv) (a) Many candidates answered this sub-part correctly. Some candidates provided a very general answer such as – it is 'collection of rainwater'.
- (b) This question was well answered by majority of the candidates.

Suggestions for teachers

- Provide more practice of reasoning-based question.
- Explain the 'type of terrain' necessary for the development of various means of irrigation.
- Discuss and explain 'two types of canals', its 'source' along with their 'merits and demerits' of both kinds of canals.
- Explain the 'methods of irrigation' and its 'distribution' using a map.
- Make students aware of the linked/ connected question (If answer of first goes wrong then the second is automatically considered incorrect).
- Do a comparative study with the help of flow chart or table for 'different means of irrigation' with the advantages and disadvantages of each.
- Guide students to mention correct definitions using key words.
- Explain 'water crisis faced by India' and the 'remedies' for the same.

MARKING SCHEME

Question 7

- | | |
|-----|---|
| (i) | <p>(a)</p> <ul style="list-style-type: none"> – India experiences a tropical monsoon type of climate, Rainfall is seasonal/ erratic/ unreliable/ irregular/ uncertain/ unevenly distributed/ sporadic. – The country has a very long growing season if water for the crop is ensured. Some crops like rice & sugarcane need more water for higher productivity of crops. <p>(b)</p> <ul style="list-style-type: none"> – Well, is simplest & cheapest source. – It is an independent source/ soft soil/ high water table. – Well can be dug at any convenient place. |
|-----|---|

	– Several chemicals are mixed in the Well/ Well need small area.
(ii)	Inundation Canal – These canals are taken out from the rivers without any regulating systems like weir and barrages at their head to regulate the flow of river. – The canals are taken out from the rivers when they are in flood and there is excess of water. – Only the lower – level regions can be irrigated by such canals. Perennial Canal – These canals are useful throughout the year as they draw their water from perennial rivers, artificial lakes. – As these canals serve throughout the year – The agricultural production in the areas of less or uncertain rainfall has increased considerably. As these canals supply water even during hot season/ dry season – Cultivation of some crops can be carried round the year.
(iii)	(a) Tank irrigation is prevalent in Deccan plateau/ Peninsular area/ Peninsular plateau widely used in Tamil Nadu, Andhra Pradesh, Maharashtra/Karnataka, Odisha, Madhya Pradesh, Rajasthan/ Uttar Pradesh/ Telangana (b) The rainwater is collected in natural pits and depressions which are easily available in peninsular. These regions are made of hard rock/ impermeable rock. Seasonal rivers/ seasonal rain (Any one point) (c) <u>Advantage of tank irrigation</u> Rainwater can be easily collected in more quantity. Tank irrigation is cheaper than other type of irrigation as the rainwater is collected in natural pits and depressions. As the water is collected and used in dry season, it uses the water table of the surrounding region. Operation in hands of farmer help in water conservation. (Any one point)

(iv)	(a) It means capturing rain where it falls or capturing the runoff in your own village or town. And taking measures to keep that water clean by not allowing polluting activities to take place in the catchment. (b) Helps replenish water levels The most effective reason why rainwater needs to be harvested is that it can be a solution to all woes of dwindling water levels. Provide drinking water. Provide irrigation water. Increase groundwater recharge. Reduce soil erosion Reduce flooding Reduce pollution from chemicals of surface and ground water Independent water supply Prevents surface runoff. (Any two points)
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Question 8

- (i) (a) Name the state of India that leads in the production of *Manganese*. [2]
- (b) What is the importance of *Manganese* for the *Iron and Steel industry*?
- (ii) Why is *conventional source of energy* **not** considered an *ecofriendly* source of energy? [2]
- (iii) (a) How is *Bhakra Nangal dam* beneficial for Punjab? [3]
- (b) Name *two* states where coal is mined in abundance.
- (c) Mention *one* advantage of using wind energy.
- (iv) Name the following: [3]
- (a) A mineral used to generate nuclear energy.
- (b) The non-conventional energy source that has the *most widespread* potential usage in India.
- (c) An offshore oil field of India.

Comments of Examiners

- (i) (a) Most of the candidates answered this question correctly.
 (b) Majority of candidates did not pay attention to question and mentioned general importance of 'manganese' instead of relating it to 'Iron and steel'.
- (ii) It was well answered by most of the candidates.
- (iii) (a) Majority of candidates answered this question correctly.
 (b) Several candidates responded this sub-part correctly.
 (c) Most of the candidates answered this sub-part of the question in a correct way.
- (iv) (a) Some candidates did not mention the 'radioactive mineral' but 'coal' instead, however, in some cases candidates mentioned the 'process of nuclear energy generation'.
 (b) It was answered correctly by the candidates.
 (c) Some candidates mentioned the 'oil field' that was onshore.

Suggestions for teachers

- Revisit and share 'mining areas/states for the minerals'.
- Encourage students to mention the 'uses of minerals' in tabular form to make learning easier.
- Discuss and revisit the advantages and disadvantages of conventional source of energy.
- Explain the benefits of multipurpose dams. Make students aware the names of states benefitted by the projects.
- Help students in building conceptual clarity. Students may be advised to read the question properly.
- Discuss advantages of non-conventional energy source.
- Make students understand the difference between 'onshore and offshore oil field'.

MARKING SCHEME

Question 8

(i)	(a) Madhya Pradesh is the largest producer of Manganese in the country. (b) <u>Used in Iron & Steel industry</u> It makes the Steel tough & rust proof/ increases strength of steel
(ii)	Most of the resources cause pollution (Burning of Coal). They might get exhausted very soon in future/ non-renewable. Large scale pollution is caused at the site of mining and place of use. Carbon emission/ global warming/ air pollution Pressure on resources Releases poisonous gases. Excessive mining leads to ecological disbalance Dust pollution <i>(Any two points)</i>

(iii)	(a) It provides cheap hydel power, helps in irrigation and soil conservation, control floods, recreation/ tourist/ attraction. (b) Jharkhand, Chhattisgarh, West Bengal, Odisha (Any two) (c) It is ecofriendly and renewable, reduces pressure on conventional sources, clean, sustainable, cost effective, no greenhouse gases (Any one)
(iv)	(a) Uranium, Thorium, Beryllium, Monazite sand, zirconium, plutonium (Any one) (b) Solar energy (c) Mumbai High, Aliabet, Bassein (Any one)

Question 9

- (i) Differentiate between *Intensive farming* and *Extensive farming*. [2]
- (ii) Mention *two* problems faced by the Indian farmers. [2]
- (iii) (a) Give *one* advantage of *transplantation method* of sowing rice. [3]
- (b) What is '*Ginning*'?
- (c) Name *any two* pulses grown in India.
- (iv) (a) What is meant by '*Retting*'? [3]
- (b) What kind of soil is required for the cultivation of tea?
- (c) Why is pruning necessary for tea plant?

Comments of Examiners

- (i) This question was well answered by the candidates.
- (ii) Majority of the candidates answered this question correctly.
- (iii) (a) Most of the candidates answered this sub-part correctly.
(b) Candidates answered this part of the question well, however, meaning of the term 'Ginning' was not explained correctly in some cases.
(c) This sub-part was answered correctly by the candidates.
- (iv) (a) Some of the candidates were not able to explain meaning of the term correctly.
(b) Most of the candidates responded this question well. Some candidates had written alluvial which was incorrect.
(c) Majority of the candidates were able to answer correctly, however, some candidates mentioned about 'rationing' which was incorrect.

Suggestions for teachers

- Explain in detail all 'types of farming practices' followed in India.
- Discuss problems of Indian farmers in class by providing examples from different areas.
- Focus on sharing methods of sowing rice along with the advantages and disadvantages of each.
- Ensure that the meaning of important terms is grasped well by the students and build the conceptual clarity.
- Conduct regular revisits.
- Encourage students to prepare table/chart to make concepts easier and clear.
- Discuss the concept of 'pruning along with its benefits'.

MARKING SCHEME

Question 9

(i)	Intensive – Multiple crops/ mixed farming/ more per hectare yield/ crop rotation small farms, many crops are grown, manure-fertilizers used, and good seeds are used, labour intensive. Followed in dense population areas. Extensive – Large farms, one or two crops are grown, machines are used, capital intensive. In less populated areas, more total yield, use less labour. <i>(Any two points of difference)</i>
(ii)	Subsistence agriculture, poor farmers, illiterate farmers, demand for food crops, challenges posed by globalization/ Seasonal rain/ dependance on rain/ lack of irrigation/ lack of credit/ poor seeds/ small holdings/ soil erosion/ lack of scientific approach/ pest & diseases/ middleman/ lack of storage facility. <i>(Any two points)</i>
(iii)	(a) Surety of crop, healthy crop, economical use of water. More production/ less wastage of seeds/ removal of weeds/ easy to harvest.

	(b) Separating seed from cotton fibre. (c) (<i>Arhar / Tur/ Tuar</i>), Gram, <i>Urad</i>, <i>Masoor</i>, Bengal gram, <i>chana</i>, <i>moong</i>, chickpea, soyabean, <i>lobia</i>, <i>rajma</i>. (<i>Any two</i>)
(iv)	(a) Retting of Jute is a process in which bundles of jute stalks are submerged in gently flowing clean water. It softens the stalk and enables fibres to be extracted and used. (b) Well drained forest soil/ Well drained mountain soil/ laterite. (c) To keep tea bush at low level to facilitate plucking and also to allow new leaves to grow & new shoots for branching laterally.

Question 10

- (i) How are the industries important for India? (*Two points*) [2]
- (ii) State the difference between *Agro based industry* and *Forest based industry*. Give *one* example for each type of industry. [2]
- (iii) (a) What has made *Mumbai* an important centre for *cotton textile* production? (*Write any two points*) [3]
- (b) Why are the *petrochemical* products in great demand?
- (iv) (a) Mention *two* problems faced by the *Sugar industry* of India. [3]
- (b) Where is TATA STEEL located?
- (c) Why is the *Silk industry* losing the market in recent times?

Comments of Examiners

- (i) This question was responded by the candidates in a correct manner.
- (ii) Majority of the candidates answered this question correctly.
- (iii) (a) This sub-part was responded well by many of the candidates.
(b) Most of the candidates answered this sub-part of the question well.
- (iv) (a) Candidates answered this sub-part in a correct way.
(b) It was answered correctly by the candidates.
(c) Candidates were not able to explain the 'impact of recent changes in market conditions' leading to 'poor performance of silk industry'.

Suggestions for teachers

- Explain in detail the 'need for industrial growth in India'.
- Guide students to use complete sentences while framing the answers.
- Share 'classification of industries' in detail with examples.
- Discuss in detail the 'factors for location of industries'.
- Encourage students to read the question prior to attempting.
- Elaborate on the 'problems of sugar industry of India'.
- Use maps while sharing the concept 'Location of Industries'.
- Connect students with the 'recent changes taking place in India' in relation to the topics covered.

MARKING SCHEME

Question 10

(i)	Industries help in making India self-reliant, provide employment, make standard of living better, increase national income, earn foreign exchange. Makes products for daily use/ makes goods & equipment used for defence. Counters regional imbalances. Help in progress of nation/ promote balance of trade.
(ii)	Agro – industries depending on agriculture for their raw material. – Cotton Textile, sugar, jute, food processing industries...etc rubber, tea, coffee, edible oil. <i>(any one example)</i> Forest – industries depending on forest for their raw material. – Paper, furniture ...etc. lac, gum, raisin, pharmaceutical, sports goods. <i>(any one example)</i>

(iii)	(a) Humid climate, availability of raw material, cheap HEP, cheap labour, port facility, market, plenty of soft water, capital, transport network (b) Cheap, cost effective, durable, attractive not dependent on agriculture, economically stable <i>(any one point)</i>
(iv)	(a) seasonal, old machines, power cut. Underutilisation of by-products/ diversion for khandsari, transport bottleneck, R M supply waver, short crushing season, poor cane, rate fixed by govt., strike & lockout. (b) It is located in Tata nagar at Jamshedpur in Jharkhand. Jamshedpur confluence of Subarnarekha & Kharkai (c) Competition from synthetic fibres, expensive, competition from other countries. Outdated powerloom/ absence of market/ not as per modern taste/ absence of skilled labour/ no testing & grading.

Question 11

- (i) Why is it important to have a well-developed means of transport in India? [2]
- (ii) “Though an expensive mode of transport, **Airways** is gaining popularity in India.” Explain giving two reasons. [2]
- (iii) (a) Why are *South Indian rivers* not ideal for inland water transport? [3]
(b) Mention *two* advantages of **waterways**.
- (iv) (a) Why are **roadways** considered more important than any other means of transport? [3]
(b) Mention two advantages of using **railways** as means of transport.

Comments of Examiners

- (i) This question was well answered by the candidates.
- (ii) Candidates answered this question correctly.
- (iii) (a) This sub-part of the question was responded well by the candidates; however, some candidates related the 'inland water transport to the coastal'.
- (b) It was answered correctly by the candidates.
- (iv) (a) Majority of the candidates answered this sub-part correctly.
- (b) Most of the candidates responded to this sub-part well.

Suggestions for teachers

- Relate a well-developed transport network to the growth of industries and economy.
- Guide students to provide reasons for a 'particular mode of transport'.
- Help students to understand the difference between 'inland' and 'coastal water transport'.
- Explain the characteristics of the rivers of both north and south India to enable students understand reasons that the 'South Indian rivers are not navigable'.
- Share both positive and negative aspects of different modes of transport.
- Discuss importance of different modes of transport in tabular form for easy learning.
- Share with students that 'no means of transport' is safer than the other. Help them to understand 'advantages and disadvantages' related to 'Safety'.

MARKING SCHEME

Question 11

(i)	For goods and raw material movement / minimise effect of natural disaster For industrial growth. / National integration/ for labour movement/ for development of economy/ for reducing regional imbalances/ for urbanisation/ development of agriculture and tourism/ for development of backward areas
(ii)	Airways are extensively used because: – Airways are fastest mode of transport/ useful during natural calamity. – Airways are free from geographical contrasts like mountain, rivers, or oceans, for example, northeast region. Saves time/ useful for perishable goods

(iii)	(a) Peninsular rivers are not ideal for navigation because the rivers are rain-fed or seasonal, they are shorter flows from uneven terrain, have number of waterfalls, because of competition among airlines price is lower (b) Advantages of waterways are: – Cheapest means of transport – Suitable for heavy and bulky goods – Fuel efficient. – Eco friendly – Less traffic – Comparatively safe <i>(Any two)</i>
(iv)	(a) – Construction cost is much lower than any other mode. Better connectivity and door to door delivery is possible which makes roadways more important than any other means of transportation. – Link between other modes – Flexible usage – For perishable commodities (b) Advantages of railways are: – Railway serves as the cheapest mode of transport for both freight and passengers over long distances. – Promote National integrity. – Suitable for long distance/ suitable for bulky goods. – Provides employment. – Reduce suffering during natural calamity. – Movement for defence personnel – Can be constructed in different terrain. <i>(Any two points)</i>

Question 12

- (i) Why should *Waste Management* be practiced in every *school*? [2]
- (ii) Mention *one* advantage and *one* disadvantage of **dumping** of waste. [2]
- (iii) Briefly answer the following: [3]
- (a) How is *segregation* of the waste helpful?
- (b) Mention *one* way in which we can *reuse* waste.
- (c) Mention *one* benefit of *Composting*.
- (iv) (a) How is *recycling* of waste helpful? Give an example of recycling of waste. [3]
- (b) Mention *any one* initiative taken by the *Government* to manage waste.

Comments of Examiners

- (i) Majority of the candidates mentioned answers that had less relevant portion.
- (ii) Most of the candidates were able to answer this question well.
- (iii) (a) This sub-part was answered correctly by the candidates; however, some candidates mentioned the ‘meaning of segregation’ instead of its ‘advantage’.
- (b) Candidates answered this sub-part in a correct manner.
- (c) Most of the candidates responded this sub-part well.
- (iv) (a) Majority of the candidates attempted this sub-part well, however, ‘use of recycling waste’ was not answered in some cases.
- (b) It was answered correctly by several candidates.

Suggestions for teachers

- Explain the ‘impact of waste accumulation’ on the ‘environment’. Comparison based questions requires both negative and positive aspects to be covered.
- Discuss ‘different methods of waste disposal’ with its advantages and disadvantages.
- Share the meaning, process, advantages, and disadvantages of ‘waste disposal methods’.
- Focus on discussing the ‘reuse of waste’.
- Proper revision of the concept must be done.
- Clearly explain the ‘concept of 3Rs’ with examples.
- Discussion on initiatives of government and community for waste management will be useful.

MARKING SCHEME

Question 12

(i)	To inculcate the importance of optimum use of resources, to analyse/ understand the disadvantage of waste in living/ non-living/ and on the environment, to learn the importance of 3Rs. <i>(Any two related points.)</i>
(ii)	Advantage - saves time, saves labour, requires less/low investment. <i>(Any one point)</i> Disadvantage – Spread of diseases and pollutants foul smell. Spoils scenic beauty/ breeding ground for mosquitoes & rodents Pollution of water (surface & ground) <i>(Any one point)</i>
(iii)	(a) Segregation leads to better and cleaner waste disposal. It encourages the 3 Rs (b) By reusing plastic bottles, cloth bags, converting waste into useful articles by DIY. – e.g., of reusing waste. (c) Benefits of Composting: – Compost improves the texture of the soil. – It aids plant growth and increases the water holding capacity of the soil. It prevents pollution/ checks soil erosion/ reduces waste accumulation. – Enhance soil nutrients. – Suppresses plant diseases/ rejuvenates poor soil by adding humus. – Converts waste into valuable fertilizer – Reduces waste accumulation.
(iv)	(a) Recycle of waste is useful for reducing disposal burden and also reducing pressure on useful resources. (e.g., Wastepaper recycled and used to make newsprint.) (b) Check degradation of land/check soil erosion, promote wasteland management, restoration of river water quality programmes, conservation of forests/ wetlands/mangroves, to make laws and acts for environment protection, etc.... Provide facilities such as dumping area, incinerators, integrated watershed management, raising awareness of 3Rs, Swachh Bharat Abhiyan <i>(Any one point)</i>

GENERAL COMMENTS

**Topics
found
difficult/
confusing by
candidates**

(a) Topics/Concepts found difficult:

- Area calculation in toposheet.
- Concept of relative height.
- Marking Satpura and Karakoram Pass on map of India.
- Reasoning questions on climate.
- Reasoning questions on soil and Natural vegetation.
- Explanation of the meaning of terms like – Ginning, Retting.
- Problems of Indian agriculture.

(b) Topics/Concepts between which candidates were confused:

- Area calculation in toposheet.
- Difference between river and stream.
- Writing of units.
- Waste management concepts such as segregation, recycling, government's contribution.

**Suggestions
for
candidates**

- Make attempt to comprehend/understand a concept prior to memorizing it.
- Please read and cover all the topics instead of selective study.
- Read the question properly before attempting the same.
- Do the practice of map regularly.
- Solve previous year's question papers.
- Prepare notes, tables/charts, and revise with its help.
- Mention the 'unit' after the answer (as asked in the question paper).
- Please write to the point short answers and use complete sentences while framing the answers.
- Mention the points in the answer as per the marks allotted for the question.
- Solve as many Multiple Choice Questions as possible for better performance.
- Write neatly and ensure to mention key words while answering.
- 'Differentiate between' questions must be answered in a tabular form.
- Take help of your teacher for any clarifications required.